

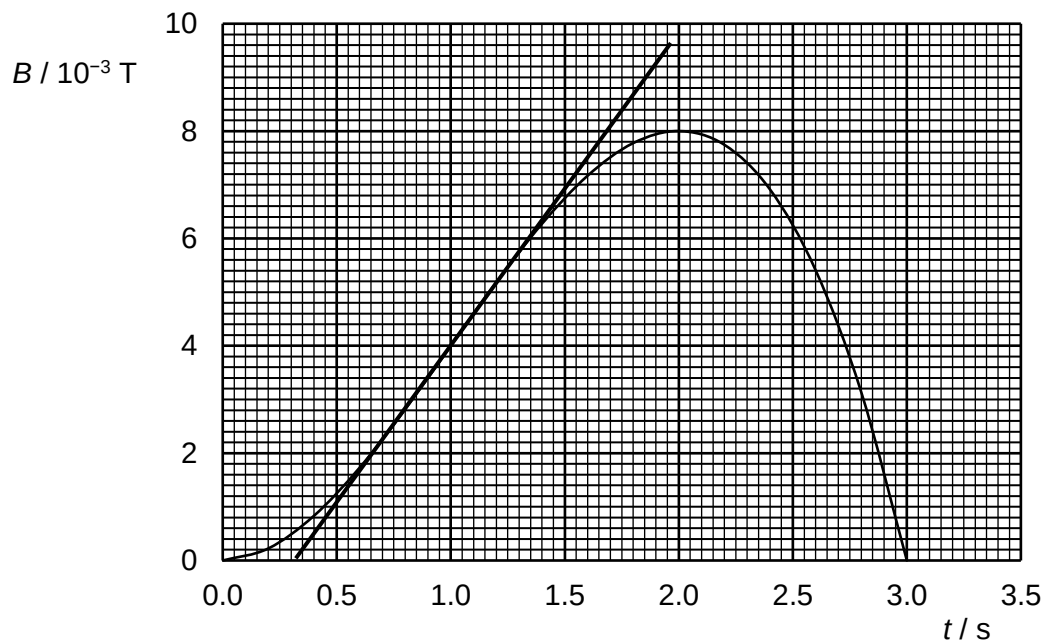
The ammeter reading will increase.

6 (a) $\Phi = BAN$

$$= (8.0 \times 10^{-3}) (\pi \times 0.17^2) \times 30$$

$$= 0.0218 \text{ Wb}$$

(b)



Since induced e.m.f. $E = -\frac{d\Phi}{dt}$, the max. e.m.f. is induced when the flux linkage Φ is changing at the highest rate, and this corresponds to when time $t = 1.0 \text{ s}$.

$$\text{Max } E = -\left(\frac{d\Phi}{dt}\right)_{\text{max}} = -AN\left(\frac{dB}{dt}\right)_{\text{max}}$$

where $\left(\frac{dB}{dt}\right)_{\max}$ is the gradient of the B vs t graph at $t = 1.0$ s

(As shown above) draw a tangent to the curve at $t = 1.0$ s.
Taking the points (1.35, 6.0) and (0.65, 2.0) on the tangent,

$$\text{gradient} = \left[\frac{6.0 - 2.0}{1.35 - 0.65} \right] \times 10^{-3} = 5.714 \times 10^{-3}$$

$$\begin{aligned} E_{\max} &= -\pi \times 0.17^2 \times 30 \times 5.714 \times 10^{-3} \\ &= -0.0156 \text{ V} \end{aligned}$$

- (c) From $t = 0$ to 2.0 s, the magnetic flux density is increasing. Thus, by Lenz's Law, the induced current produces an induced magnetic field upwards to oppose the increase in the magnetic flux in the coil.
By right hand grip rule, the direction of the induced current is anti-clockwise.

(d)

